UNIT 1 · TOPIC 6(E)

Baling Equipment

Syllabus topic: Equipment for harvesting, threshing, mowing, chaff cutting and baling

Rectangular (square) baler (Kepner ch. 14)

1. The **pickup** lifts the windrow.
2. The **auger/packer** feeds the bale chamber.
3. The **plunger** (crank-driven, ~**65–90 strokes/min**) compresses each charge in the bale chamber.
4. The **star/metering wheel** measures bale length.
5. The **knotter** (Deering/Appleby type — billhook, twine disc, wiper arm) ties the twine when the bale reaches the set length.

- Tension rails adjust bale density; a bale weighs **~35–125 kg**.

Round baler

- Belts/rollers roll the windrow into a cylindrical bale (variable or fixed chamber), wrapped with net or twine.
- It weathers better — rain sheds off.

Straw balers with a pickup, working behind combines, are central to residue management in NW India.

Hay making, step by step

1. Mow the crop.
2. Condition it — crusher/crimper rolls speed drying.
3. Rake it into windrows.
4. Bale at **18–22%** moisture (loose stack ~**25%**).
5. Store baled hay at **≤ ~20%** moisture to avoid heating and mould.

Practice MCQs

Q57. In a rectangular baler, twine is tied by the:

(a) Plunger
(b) Knotter (billhook mechanism)
(c) Pickup
(d) Star wheel

Answers

Q57 — (b) Knotter (billhook mechanism)